

Pablo Leyva

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EDUCATION

New Jersey Institute of Technology

B.S. in Applied Mathematics (Statistics) & Computer Science (AI Concentration)

Newark, NJ

Aug 2023 – May 2027

- **Relevant Coursework:** Reinforcement Learning, Deep Learning, Probability, Linear Algebra, Convex Optimization

EXPERIENCE

Apple

May 2026 – Present

Enterprise GenAI Product Research Intern

Cupertino, CA

- Built **model-based tools** that curate and score **RL post-training data** for a continually-learning SLM; own the team's data-quality workstream studying how curation drives out-of-distribution generalization on agentic Apple Wallet and Apple Pay workflows
- Designing **RL environments with verifiable rewards** (program-analysis-style oracles for payment-flow correctness); implementing **GRPO/PPO** on-policy training with vLLM rollouts and a reward model trained on preference data

Undergrad Research and Innovation (URI) - Agentic Reasoning SLM

Sep 2025 – Present

Research Fellow

Newark, NJ

- Training a **3B-parameter reasoning SLM** via **SFT** → **DPO** → **RL fine-tuning** with multi-GPU distributed training (FSDP) on math-proof and logic-puzzle datasets; targeting a public preprint on one-step reasoning generalization in small language models
- Constructed a **synthetic RL environment** with programmatically-checkable rewards for mathematical proof search; ran **ablation studies** across reward-shaping strategies and data-mixing ratios to isolate sources of reasoning gain
- Ran **linear probes** on hidden activations and **safety evaluations** (refusal consistency, reward hacking, out-of-distribution robustness), benchmarking the SLM against larger LLM baselines on lm-eval-harness

Apple

May 2025 – August 2025

AI/ML Product Engineering Intern

Cupertino, CA

- Built an **agent orchestration system** integrating the Apple Foundation Model (AFM) with Model Context Protocol (MCP); collected **trajectory-level preference data** across 10k+ rollouts to bootstrap a reward-model prototype
- Built the team's **LLM evaluation pipeline** benchmarking agent trajectories (tool-use success, groundedness, hallucination rate) against Claude and GPT baselines; results directly informed fine-tuning priorities
- Designed **tool selection and execution** flows with LangGraph using TouchID human-in-the-loop validation; presented the prototype at the team's end-of-internship review (org director present)

Caterpillar Inc.

May 2024 – August 2024

Software Engineering Intern

Chicago, IL

- Fine-tuned **transformer LLMs** (BERT, GPT-3.5) with **LoRA/QLoRA** on a 500K-record corpus; ran ablation studies on data mixing and learning-rate schedules, reaching 89% classification accuracy
- Built an **end-to-end data pipeline** with Python and Apache Airflow ingesting 500K+ daily records; served model predictions via **FastAPI microservices** at sub-100ms latency for 10K+ daily requests

PROJECTS

Self-Evolving Agent Harness | *PyTorch, TRL, Gymnasium, vLLM, LoRA, Weights & Biases* | **Python**

- Built the first **agent harness for non-technical users** that auto-generates **personalized tools** from natural-language needs and curates on-policy trajectories to **fine-tune a per-user policy (LoRA)**, internalizing recurring context so the agent no longer re-reads static memory files, cutting per-turn tokens by ~60%

Mechanistic Interpretability of Reward Models | *PyTorch, TransformerLens, sae_lens, HuggingFace, Anthropic hh-rlhf* | **Python**

- Trained **sparse autoencoders** on mid-layer activations of open-source reward models (Skywork-Reward, DeBERTa-RM) over **preference-pair datasets** (hh-rlhf, UltraFeedback) to extract interpretable features driving the scalar reward
- Ranked SAE features by **causal influence on reward score** and flagged candidate **reward-hacking proxies** (response length, hedging, markdown formatting, sycophantic apology) that correlate with RM score but not with human preference
- Implemented **activation steering** mitigation that ablates hacky feature directions at inference, recovering ~8 pts of preference accuracy on held-out adversarial pairs; writeup in draft for Alignment Forum / arXiv

COMMUNITY OUTREACH

President, Vice President - External Affairs, Event Coordinator

October 2023 – Present

Society of Hispanic Professional Engineers (SHPE)

Newark, NJ

- Leading a team of 20 to organize professional events for 300+ students; grew external company partnerships 6x and averaged 200% growth in weekly member engagement
- Built and shipped the chapter's first **iOS app** in 27 days with a team of 5 engineers, growing it to **200+ users** in its first month

TECHNICAL SKILLS

Languages : Python (fluent), C++, CUDA, Java

ML Frameworks : PyTorch, JAX, Hugging Face Transformers, TRL, DeepSpeed, FSDP, vLLM, Ray

Reinforcement Learning : PPO, GRPO, DPO, RLHF, reward modeling, Gymnasium, RL environment design, preference data

LLM Training & Evaluation : SFT, LoRA/QLoRA, fine-tuning, distillation, synthetic data, lm-eval-harness, safety evaluations, ablation studies

Interpretability & Alignment : mechanistic interpretability, sparse autoencoders, activation steering, scaling laws, reward hacking